

1 **Immune gene expression across blood, cerebrospinal fluid and**
2 **nerve in inflammatory neuropathies**

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14 **Abstract**

15 Inflammatory neuropathies are commonly studied in a single biospecimen, limiting
16 understanding of systemic and tissue immune responses. We examined complementary
17 evidence from blood, cerebrospinal fluid and peripheral nerve in Guillain–Barré syndrome
18 and chronic inflammatory demyelinating polyradiculoneuropathy. Three acute blood
19 transcriptomic cohorts were analysed using gene module scores and models with random
20 effects. Published cerebrospinal fluid protein coefficients were combined using a model with
21 a fixed effect. A sural nerve atlas was analysed at donor level, focusing on nine donors with
22 chronic inflammatory demyelinating polyradiculoneuropathy and 11 with chronic idiopathic
23 axonal polyneuropathy. Macrophage findings underwent exploratory demographic,
24 composition and annotation sensitivity analyses. Published genetic candidates were mapped to
25 nerve cell states. CXCL8 recruitment and complement modules had positive effects in all
26 three blood cohorts, but pooled confidence intervals included zero: Hedges g 1.08 (95%
27 confidence interval -0.63 to 2.78) and 1.11 (-1.73 to 3.95), respectively. Nine of ten selected
28 cerebrospinal fluid proteins had positive coefficients in both cohorts and positive pooled
29 effects after correction across ten tests. CD5 was discordant and its pooled effect was not
30 significant. In nerve, the macrophage Fc receptor score was lower in the inflammatory
31 neuropathy group after correction across eight modules within macrophages ($q = 0.0087$), but
32 not across all 80 eligible module comparisons ($q = 0.087$). The association persisted after age
33 and centre adjustment, but weakened when macrophage composition was included. No
34 eligible refined macrophage subtype met its correction threshold. Intermodule and disability
35 correlations also did not survive correction. Expression localization placed *CDH4*
36 predominantly in Schwann cell states and *GNG7* in B cells, with expression in additional cell
37 groups. These independently sampled datasets describe complementary expression patterns
38 and an exploratory macrophage association that remains sensitive to analytical scope and

39 composition. Independent cohorts and functional experiments are needed to establish
40 biological mechanisms.
41 Keywords: neuroimmunology, endoneurial endothelium, pseudobulk analysis, proteomics,
42 genetic susceptibility

43 **Abbreviated summary**

44 Feng et al. examine blood, cerebrospinal fluid and nerve expression in inflammatory
45 neuropathies. A macrophage Fc receptor association persists after demographic adjustment
46 but weakens after composition adjustment and broader multiple testing correction. These
47 findings refine hypotheses for independent validation without establishing communication
48 between compartments.

49 **Abbreviations**

50 AIDP, acute inflammatory demyelinating polyradiculoneuropathy. BNB, blood nerve barrier.
51 CIAP, chronic idiopathic axonal polyneuropathy. CIDP, chronic inflammatory demyelinating
52 polyradiculoneuropathy. CPM, counts per million. FDR, false discovery rate. GBS, Guillain–
53 Barré syndrome. INCAT, Inflammatory Neuropathy Cause and Treatment. PBMC, peripheral
54 blood mononuclear cell.

55

56 **Introduction**

57 Guillain–Barré syndrome (GBS) and chronic inflammatory demyelinating
58 polyradiculoneuropathy (CIDP) differ in their clinical course and diagnostic criteria.^{1,2,3} Injury
59 associated with antibodies, complement deposition and demyelination mediated by
60 macrophages contribute to inflammatory neuropathy, with their relative importance varying
61 by disease and subtype.^{4,5,6,7,8,9} Peripheral nerve cell atlases and experimental injury studies
62 also identify distinct immune and Schwann cell responses.^{10,11,12,13} Leukemia inhibitory factor
63 (LIF) can influence Schwann cell survival, proliferation and migration in experimental
64 systems.^{14,15} These observations motivate examination of gene expression related to immune
65 responses and repair across tissues, although a shared expression pattern does not establish a
66 shared mechanism.

67 The blood nerve barrier (BNB) regulates exchange between the circulation and the
68 endoneurium. Endoneurial endothelial junctions and associated mural cells contribute to its
69 specialized properties.^{16,17,18} Experimental studies implicate leukocyte integrins, endothelial
70 adhesion molecules and chemokines in trafficking across this interface.^{19,20,21} Altered
71 junctional protein expression has also been described in CIDP nerve.²² The BNB is therefore a
72 plausible site at which circulating immune responses interact with tissue injury and repair.

73 Blood transcriptomic studies characterize systemic immune expression in GBS across whole
74 blood, leukocytes and peripheral blood mononuclear cells (PBMCs).^{23,24,25} CSF proteomics
75 and peptidomics provide complementary measurements of inflammation and molecular
76 injury, while nuclear transcriptomic data from nerve localize expression to annotated cell
77 states.^{26,27} Other studies report monocyte changes in acute inflammatory demyelinating
78 polyradiculoneuropathy (AIDP), peripheral immune alterations in CIDP and transcriptomic
79 associations with treatment response.^{28,29,30} Experimental evidence that patient
80 immunoglobulin can disrupt barrier properties further motivates investigation of the BNB.³¹

81 Human genetics supplies an additional route to candidate selection.³² However, findings from
82 different diseases, sample types and cohorts cannot by themselves establish communication
83 between compartments.

84 We aimed to assess the consistency of acute GBS blood gene modules across cohorts,
85 synthesize selected published CSF evidence, and localize related immune and repair programs
86 in CIDP nerve. We also mapped published CIDP genetic candidates to nerve cell states.
87 Integration was descriptive and retained the distinct sampling units and evidence levels of
88 each dataset.

89 **Materials and methods**

90 **Study design and datasets**

91 The study integrated human data from blood, CSF, peripheral nerve and genetic studies.
92 GSE211225 contained six whole blood expression profiles from acute GBS, ten from the
93 postacute phase and six from controls.²³ GSE31014 provided leukocyte microarrays from 7
94 patients with GBS and 7 controls.²⁴ PRJNA1293757 provided RNA sequencing data for
95 individual PBMCs collected from 3 untreated patients with AIDP and 2 controls.²⁵ These
96 three datasets were used to assess the consistency of blood findings across cohorts.
97 Published findings associated with PXD002911 provided CSF protein and endogenous
98 peptide identifications by diagnostic group.²⁶ GSE285983 contained 365,708 nuclei from 37
99 sural nerves, including nine donors with CIDP and 11 with chronic idiopathic axonal
100 polyneuropathy (CIAP).²⁷ The published GSE107574 BNB transcriptome supplied biological
101 context. Its raw expression data were not independently reanalysed here.¹⁷ Genetic candidates
102 were selected from the analyses of CIDP reported by Du et al., which included association
103 testing across the genome, Mendelian randomization and colocalization.³² Additional CSF
104 reports contributed descriptive evidence on complement, quantitative proteomics and
105 inflammatory proteins.^{33,34,35} We extracted adjusted CSF coefficients from the independent

106 Olink discovery and replication cohorts reported by Kmezic et al.³⁶ Published histological and
107 proteomic findings associated with PXD056286 provided tissue context from 55 CIDP nerve
108 biopsies.³⁷ Original resource names, accession identifiers, filenames and access links are
109 consolidated in Supplementary Data 1.

110 **Definition and scoring of biological programs**

111 Seven blood gene modules covered CXCL8 recruitment through CXCR1 and CXCR2, LIF
112 and oncostatin M (OSM) signalling through gp130, complement, Fc receptors, interferon
113 signalling through JAK and STAT, inflammatory monocytes and transendothelial migration.
114 Eight nerve modules covered BNB identity, leukocyte transmigration, CXCL8 recruitment,
115 LIF signalling, complement, Fc receptors, macrophage state and Schwann cell myelin repair.
116 Gene sets were specified in the analysis scripts. The study was not prospectively registered.
117 Module scores summarize transcription of selected genes and were not interpreted as direct
118 measurements of protein activity or cellular function. CSF features were assigned to
119 biological categories using annotations from the source studies.

120 **Analysis and synthesis of acute GBS blood cohorts**

121 Within each blood cohort, gene expression was standardized across participants and averaged
122 over available genes to obtain module scores. Differences between acute GBS and control
123 groups were quantified using Hedges g. Exact label permutation tests assessed differences in
124 mean scores, with P values calculated from both tails. Correction using the Benjamini and
125 Hochberg procedure was applied across seven modules within each cohort comparison.³⁸
126 Effects from whole blood, leukocytes and monocyte pseudobulk were combined using
127 exploratory models with random effects. Variance between studies was estimated using the
128 method of DerSimonian and Laird. Intervals calculated using the modified Hartung and
129 Knapp method used a residual variance scale bounded below by one and a t distribution with
130 degrees of freedom equal to the number of cohorts minus one. Heterogeneity was summarized

131 using τ^2 and I^2 .^{39,40} We repeated each synthesis after omitting one cohort to assess direction
132 and uncertainty using the two remaining cohorts. Differences in assay platform, cellular
133 composition and available module genes limit interpretation of a common pooled effect.
134 Postacute profiles were retained in the original preprocessing, but no inference about
135 differences between disease phases is presented here because independence of donors across
136 phases could not be verified from the archived design.

137 **Synthesis of published GBS CSF proteomic evidence**

138 CSF evidence was assembled from selected reports rather than a systematic literature review.
139 For the quantitative synthesis, all ten CSF proteins other than IL8 with coefficients and
140 standard errors in both cohorts of Table 4B in Kmezic et al. were eligible, irrespective of
141 effect direction or statistical significance.³⁶ This corrected an earlier extraction that omitted
142 IL6 and CD5. The source regressions adjusted for age, sex and sample handling. Pooled
143 estimates were calculated using a model with a fixed effect and weighting by inverse
144 variance. The 95% confidence intervals used a normal approximation, and correction using
145 the Benjamini and Hochberg procedure covered ten pooled tests. Estimates calculated using
146 the method of DerSimonian and Laird, Q and I^2 were reported as heterogeneity assessments.
147 The normal approximation does not reproduce the tests used in the source regressions for
148 small samples, and pooled significance does not establish significance in both cohorts. IL8,
149 the source study's separately reported primary biomarker in Table 3, was retained as
150 contextual evidence. The selected table does not contain the full assayed proteome. Published
151 identifications and annotations from PXD002911 and three other reports were recorded in a
152 descriptive evidence map.^{26,33,34,35} Counts of reports and platforms describe coverage of these
153 four studies, not effect magnitude or statistical strength. No CSF matrix containing data for
154 individual participants was reconstructed or reanalysed.

155 **Pseudobulk analysis by donor and cell state in CIDP nerve**

156 Nuclei from GSE285983 were grouped by donor and annotated cell state. Focused groups
157 included macrophages, T and natural killer cells, B cells, granulocytes, endoneurial
158 endothelial cells, other endothelial cells, pericytes, perineurium, endoneurial stroma and
159 Schwann cells in myelinating, nonmyelinating and repair and damage states. A cell group was
160 eligible for a given donor if it contained at least 20 nuclei. Counts were summed within each
161 group and transformed as $\log_2(\text{CPM} + 0.5)$, using counts across all genes for the library size.
162 Donors were the biological replicates for comparisons between CIDP and CIAP.^{41,42,43}

163 **Gene module scores and cellular composition**

164 Within each cell group, each gene was standardized across eligible donors from all diagnostic
165 categories. Module scores were means of the available gene z scores. CIDP and CIAP were
166 compared using the U test of Mann and Whitney with two tails. Effect estimates were
167 calculated as mean CIDP scores minus mean CIAP scores. The original correction using the
168 Benjamini and Hochberg procedure covered eight modules within each cell group. We
169 additionally evaluated correction across all 80 eligible module comparisons for the CIDP
170 versus CIAP contrast. Donor cell fractions were compared separately, with correction across
171 cell groups. Because fractions are compositional, these comparisons describe relative
172 representation.⁴⁴

173 The macrophage Fc module contained 11 measured genes from 12 specified genes because
174 FCGR2C was absent. We examined individual genes with correction across 11 tests and four
175 functional summaries with a separate correction. Sensitivity analyses restricted the
176 comparison to male donors, standardized expression within the 20 case and comparator
177 donors, adjusted for age and centre, and additionally included sex. Linear model intervals
178 used HC3 covariance and residual degrees of freedom. An exact permutation test preserved

179 the observed number of CIDP donors within each centre. Details are provided in
180 Supplementary Methods.
181 Original macrophage annotations were joined to the refined immune annotations by nucleus
182 barcode. We retained unmatched and reassigned categories when describing composition. A
183 sensitivity model added the first two principal components of donor composition, represented
184 as centred log ratios, to age and centre. Raw count matrices for all 20 CIDP and CIAP donors
185 were reaggregated under alternative macrophage definitions. Refined subtype tests required at
186 least 20 nuclei per donor and at least three donors per diagnosis. Correction covered all nine
187 eligible subtype tests. These analyses assess sensitivity to annotation and composition without
188 formally separating biological mediation from confounding.

189 **Coordination of tissue programs and associations with disability**

190 Five module scores were paired by donor: BNB endothelial LIF, endothelial leukocyte
191 transmigration, macrophage complement, macrophage Fc receptors and myelin repair in
192 Schwann cells in the repair and damage state. Spearman correlations were calculated for their
193 ten pairwise combinations across nine CIDP donors. Fifteen combinations of cell group and
194 module were examined for associations with Inflammatory Neuropathy Cause and Treatment
195 (INCAT) disability scores. Correction using the Benjamini and Hochberg procedure was
196 applied separately to the ten intermodule and 15 clinical correlations. These analyses describe
197 covariation. Interactions between ligands and receptors were not inferred.

198 **Integration across compartments and assessment of robustness**

199 Integration across compartments focused on recurrent biological programs rather than
200 individual genes. A program was included in the integrated map if it showed a consistent
201 direction across blood cohorts, had a corresponding published protein or peptide annotation in
202 CSF, or was localized to a relevant cell state in nerve. The map distinguished effects
203 calculated from blood cohorts, new synthesis of published CSF coefficients, descriptive CSF

204 evidence from the literature and nerve localization assessed for individual donors. These
205 forms of evidence were displayed separately without combining them into a single numerical
206 score. Because blood, CSF and nerve samples came from independent cohorts,
207 correspondence across compartments was not interpreted as evidence of temporal progression,
208 cell migration or causal signaling.

209 **Cellular localization of published CIDP genetic candidates**

210 We selected *CDH4*, *DIRAS1*, *GNG7* and *SLC39A3* from the published CIDP study of 516
211 cases and 403,545 controls.³² The *CDH4* locus reported in women and the Mendelian
212 randomization and colocalization estimates for the other three genes were treated as prior
213 genetic evidence. The new analyses examined the cellular expression of these candidates,
214 differences between CIDP and CIAP, and the stability of localization across donors. Raw
215 CellBender counts from GSE285983 were normalized as $\log_2(\text{CP10k}+1)$ for each nucleus and
216 summarized by donor and grouped nerve cell type. We reported the fraction of nuclei with
217 detectable expression and mean expression for each donor. Uncertainty in localization was
218 assessed using 5,000 bootstrap resamples of CIDP donors for each combination of gene and
219 cell type. CIDP and CIAP donors were compared using the U test of Mann and Whitney with
220 two tails, followed by correction using the Benjamini and Hochberg procedure within each
221 cell group. Complete summary statistics from the custom FinnGen GWAS were available
222 only on request, so independent GWAS, MAGMA, Mendelian randomization, colocalization
223 and TWAS analyses were not performed.

224 **Statistical analysis**

225 Participants and nerve donors were the independent biological samples. Cells from the same
226 donor were not treated as independent observations. All P values were calculated from tests
227 with two tails. P values adjusted using the Benjamini and Hochberg procedure are denoted q.
228 A threshold of $q < 0.05$ was used within each specified testing family. New Fc sensitivity

analyses were exploratory and were specified after the original association was known. Their testing families do not provide independent confirmation of that association. Scripts accompanying this revision document the corrected CSF extraction, reproduce the Fc summaries and reaggregate the 20 nerve count matrices for annotation checks. Other original matrix processing and genetic bootstrapping were retained from the preceding analysis.

Ethical considerations

This study used secondary data from public resources and published reports. No participants were recruited and no new specimens were collected. The source studies reported their own ethics approvals and consent procedures.

Results

Study resources and analytical roles

GSE211225 characterized acute expression in whole blood, while GSE31014 and PRJNA1293757 assessed directional agreement in leukocytes and monocytes. Published CSF studies supplied evidence from protein measurements, and GSE285983 localized selected genes and modules in nerve. The published BNB reference informed interpretation of the endothelial context (Fig. 1 and Table 1).

Acute blood expression and consistency across cohorts

GSE211225 showed a coordinated transcriptional response in whole blood during acute GBS. Compared with healthy controls, patients had higher scores for CXCL8 recruitment through CXCR1 and CXCR2 (Hedges $g = 1.25$, $q = 0.045$), LIF and OSM signaling through gp130 ($g = 2.19$, $q = 0.0076$), complement ($g = 2.49$, $q = 0.0076$), Fc receptors ($g = 1.36$, $q = 0.038$), interferon signaling through JAK and STAT ($g = 1.07$, $q = 0.045$) and inflammatory monocytes ($g = 1.93$, $q = 0.015$). The transendothelial migration module score was also higher

252 ($g = 0.92$), although q was 0.126. These results identify increased expression of genes
 253 involved in chemotaxis and humoral effector responses.
 254 The independent cohorts helped identify which responses were consistent across datasets.
 255 GSE31014 showed positive effects for six of the seven programs. Scores for CXCL8
 256 signaling through CXCR1 and CXCR2, complement, Fc receptors, transendothelial migration,
 257 interferon and inflammatory monocytes all changed in the same direction as in whole blood.
 258 In PRJNA1293757 monocytes, CXCL8 signaling through CXCR1 and CXCR2 had the
 259 largest positive effect ($g = 1.97$), and the complement effect was also positive. The remaining
 260 monocyte effects varied in direction, providing clearer support for localization of the
 261 recruitment response than for reproduction of the full whole blood profile.
 262 CXCL8 recruitment and complement effects were positive in all three datasets. The pooled
 263 Hedges g was 1.08 for CXCL8 recruitment (95% CI -0.63 to 2.78 , $P = 0.114$ using the
 264 modified Hartung and Knapp method, $I^2 = 0\%$) and 1.11 for complement (95% CI -1.73 to
 265 3.95 , $P = 0.234$, $I^2 = 57.2\%$). None of the seven pooled module effects reached $P < 0.05$. Four
 266 other modules were positive in two cohorts, whereas the module for LIF and OSM signalling
 267 was positive only in whole blood. Directional agreement therefore identified exploratory
 268 signals, with substantial uncertainty in their pooled magnitude (Fig. 2 and Table 2).
 269 After successive omission of each cohort, CXCL8 recruitment summary estimates remained
 270 positive ($g = 0.97$ to 1.41), as did complement estimates ($g = 0.48$ to 1.55). Both individual
 271 effects in each retained pair were also positive. Other modules showed at least one discordant
 272 cohort. Intervals based on two studies were wide. Persistence of direction did not provide
 273 additional evidence of statistical significance.
 274 Thus, six blood modules met the criterion for correction within the cohort in GSE211225,
 275 while only CXCL8 recruitment and complement had positive effects across all three cohorts.

276 This distinction between a result from one cohort and consistency in direction across cohorts
277 guided the subsequent descriptive integration.

278 **Heterogeneity across blood sample types**

279 I^2 estimates were 0% for CXCL8 recruitment and interferon signalling through JAK and
280 STAT, 1.1% for transendothelial migration, 81.2% for LIF and OSM, 72.7% for Fc receptors,
281 57.2% for complement and 49.5% for inflammatory monocytes. With only three cohorts,
282 these estimates are themselves uncertain. Variation may reflect assay platform, cell
283 composition, gene coverage or clinical differences. The present design cannot separate these
284 explanations.

285 These differences also help interpret the blood findings. Recruitment signals were detectable
286 in both mixed blood populations and monocytes, whereas the magnitude of LIF and OSM, Fc
287 receptor and monocyte state signals varied with the cell population sampled. We therefore
288 used synthesis across cohorts to identify recurrent directions while retaining estimates from
289 individual cohorts to describe the broader immune response measured by each platform.

290 **Corrected synthesis of published CSF protein coefficients**

291 Ten CSF proteins other than IL8 had paired adjusted coefficients in the selected source
292 table.³⁶ Nine proteins, namely SELE, IL2RA, CCL3, CR2, CD1C, THBD, NRP1, CD38 and
293 IL6, had positive effects in both cohorts. Their pooled estimates from the model with a fixed
294 effect had $q < 0.05$ after correction across ten tests. CD5 was discordant: its discovery
295 coefficient was -0.0882 and its replication coefficient was 1.51 . Its pooled coefficient was
296 0.0024 (95% CI -0.237 to 0.242 , $q = 0.984$). Heterogeneity was high for IL2RA ($I^2 = 94.9\%$),
297 CD5 (89.0%), IL6 (80.7%) and THBD (79.5%). These pooled results must be distinguished
298 from the source study's finding that only SELE and IL2RA remained significant in both
299 cohorts after its own adjustment for sample handling and correction for multiple testing
300 among the proteins other than IL8 (Fig. 3A and B).

301 The descriptive evidence map provided context from four complementary CSF studies. The
302 report linked to PXD002911 contributed protein and peptide identifications and functional
303 annotations.²⁶ D'Aguanno et al. contributed complement and inflammatory protein findings,³³
304 Ding et al. contributed protein handling and neural structure findings,³⁴ and Sun et al.
305 contributed inflammatory protein and routine CSF measurements.³⁵ Across these selected
306 reports, local inflammation was represented in three studies and protein accumulation in two.
307 Structural proteolysis, complement, and protein handling and neural structure were each
308 represented in one study (Fig. 3C). These counts summarize the coverage of published
309 evidence alongside the new synthesis of Olink coefficients.

310 **Gene expression related to immune responses and repair in CIDP nerve**

311 The sural nerve atlas identified the cells expressing the integrated programs within peripheral
312 nerve. Macrophages expressed C3, C3AR1 and FCGR2A, while endoneurial endothelial cells
313 expressed CLDN5, ICAM1, VCAM1, LIFR, IL6ST and OSMR. Schwann cell states
314 expressed gp130 receptor components together with genes involved in myelin, stress and
315 repair. These expression patterns provide cellular context for programs involving humoral
316 responses, endothelium and Schwann cells (Fig. 4A).
317 CLDN5 and genes associated with adhesion in the nerve atlas were interpreted in the context
318 of the published human BNB transcriptome.¹⁷ That reference supports the biological relevance
319 of endoneurial endothelium, but it was not used as an independently reanalysed validation
320 cohort.

321 The clearest difference between CIDP and CIAP was a lower macrophage Fc receptor gene
322 module score in CIDP (difference in mean z scores = -0.673 , $P = 0.0011$, $q = 0.0087$).
323 Macrophage module scores for leukocyte transmigration (difference in mean z scores =
324 -0.345 , $P = 0.0185$, $q = 0.074$) and macrophage state (difference in mean z scores = -0.368 , P
325 = 0.0334 , $q = 0.089$) were also lower, but neither met the threshold after correction for

multiple testing. In the perineurial cell group, scores for the macrophage state gene module (difference in mean z scores = 0.758, $P = 0.0227$) and complement module (difference in mean z scores = 0.348, $P = 0.0334$) were higher, with a q value of 0.109 for both. The BNB endothelial identity module score was nominally lower in CIDP (difference in mean z scores = -0.404 , $P = 0.0276$, $q = 0.221$). These estimates describe gene module expression within the annotated cell groups.

The Fc module association met the original correction threshold within macrophages, but its q value increased to 0.087 when correction covered all 80 eligible module comparisons. All nine CIDP donors were male, compared with seven of 11 CIAP donors. The direction persisted among male donors (difference in mean z scores = -0.687 , $P = 0.0007$). Adjustment for age and centre gave an estimated difference of -0.690 (95% CI -1.040 to -0.339 , $P = 0.0008$). The exact test with permutation within centres gave $P = 0.0048$. Estimates remained negative after separate omission of individual donors, centres and genes. These sensitivity analyses support consistency of the observed direction within this dataset (Supplementary Fig. 3).

Four measured genes, FCGR2A, FCER1G, TYROBP and HCK, had lower expression with $q < 0.05$ within the 11 gene component analysis. FCGR2A showed no significant difference ($q = 0.713$), and FCGR1A had a positive estimate that did not meet the correction threshold ($q = 0.148$). The signalling component summary was lower ($q = 0.039$), while the activating receptor, inhibitory receptor and neonatal Fc receptor summaries did not meet their correction threshold. The composite score therefore does not indicate uniform suppression of Fc receptors.

After adding two macrophage composition components to age and centre, the estimated difference was -0.310 (95% CI -1.038 to 0.419 , $P = 0.375$). The refined union of macrophage states retained a lower score ($P = 0.0062$), but none of the nine eligible individual refined

351 subtypes met $q < 0.05$. None of the 32 refined annotation fractions met its correction
 352 threshold. These findings leave uncertainty about the relative contributions of state
 353 composition and expression within individual states (Supplementary Fig. 4).
 354 No cell fraction differed between CIDP and CIAP after correction across cell groups. The
 355 largest numerical increases were in nonmyelinating Schwann cells (+0.064), endoneurial
 356 stroma (+0.027) and myelinating Schwann cells (+0.013), each with $q = 0.515$. These results
 357 provide limited evidence for a compositional difference and should not be interpreted as
 358 equivalence between diagnoses.
 359 Correlations calculated across the nine CIDP donors provided clinical context for the tissue
 360 findings. The myelin repair module score in Schwann cells in the repair and damage state
 361 showed the strongest association with INCAT disability ($\rho = 0.840$, $P = 0.0046$, $q = 0.069$).
 362 The BNB endothelial LIF module score showed a weaker positive association ($\rho = 0.602$, $P =$
 363 0.086). Neither association met the threshold after correction for multiple testing. They
 364 therefore describe exploratory relationships between gene expression and neurological
 365 impairment.
 366 Thus, the macrophage Fc score identifies an exploratory association whose interpretation
 367 depends on the testing family and macrophage composition. The strongest observed disability
 368 correlation involved the Schwann cell state associated with repair and damage, but remained
 369 exploratory after correction.
 370 Published tissue observations provided complementary evidence on the BNB. Stascheit et al.
 371 reported C5b-9 deposition on endoneurial capillaries in 52 of 55 biopsies from patients with
 372 CIDP or its variants.³⁷ From that published count, we calculated a proportion of 94.5% and a
 373 Wilson 95% CI of 85.1% to 98.1%. The accompanying observations of macrophages
 374 expressing CD68, T cells expressing CD8 and increased C3 and C6 abundance were taken
 375 from the source report. This evidence provides histological and proteomic context for the

376 nerve findings, without constituting a new analysis of biopsy specimens or replication in a
377 second dataset of nuclear transcriptomes (Supplementary Fig. 2).

378 **Exploratory correlations among nerve modules**

379 Across nine CIDP donors, BNB endothelial LIF scores correlated with endothelial leukocyte
380 transmigration ($\rho = 0.667$, $P = 0.0499$, $q = 0.166$) and myelin repair scores in Schwann cells
381 in the repair and damage state ($\rho = 0.783$, $P = 0.0125$, $q = 0.125$). Endothelial transmigration
382 also correlated with macrophage complement ($\rho = 0.683$, $P = 0.0424$, $q = 0.166$). None of the
383 ten intermodule correlations survived correction. These estimates generate hypotheses about
384 covariation across donors without demonstrating signalling between cell types.

385 The evidence map placed these tissue observations alongside blood and CSF findings.

386 CXCL8 recruitment and complement had directionally consistent blood effects and
387 corresponding published CSF annotations. The Fc receptor score was higher in acute GBS
388 whole blood than in healthy controls, but lower in CIDP nerve macrophages than in CIAP
389 macrophages. Because diseases, cell populations and comparators differ, this contrast cannot
390 establish a reversal between compartments within the same disease. Endothelial adhesion and
391 Schwann cell repair genes supplied tissue context, while CSF reports contributed evidence of
392 protein accumulation and proteolysis (Fig. 5 and Table 3).

393 **Cellular expression of published CIDP genetic candidates**

394 Four candidates from the published study of Du et al. were selected for nerve cell
395 localization.³² *CDH4* was supported by the locus reported in women at 20q13.33, while
396 *DIRAS1*, *GNG7* and *SLC39A3* were selected from the reported Mendelian randomization and
397 colocalization findings. Figure 5C summarizes these published estimates as the basis for
398 candidate selection.

399 In GSE285983 donor summaries, *CDH4* expression was highest in nonmyelinating Schwann
400 cells and Schwann cells in the repair and damage state. *GNG7* expression was strongest in B

401 cells, with additional expression in macrophages and BNB endothelial cells. *SLC39A3* was
402 expressed at low levels, with its highest mean in BNB endothelium, and *DIRAS1* was sparsely
403 expressed. Donor bootstrap intervals described uncertainty around these means for each cell
404 group (Supplementary Fig. 1). They were not tests of specificity for one cell type. None of the
405 displayed comparisons of candidate expression between CIDP and CIAP survived correction.
406 These findings describe cellular expression contexts without identifying a biomarker of
407 disease state or a causal effector cell (Fig. 5).

408 **Discussion**

409 This study assembled complementary blood, CSF and peripheral nerve evidence in GBS and
410 CIDP. The principal observations were directionally consistent but statistically uncertain
411 blood effects, a synthesis of selected published CSF proteins, and cellular localization of
412 expression related to inflammation and repair. A lower macrophage Fc score in CIDP than in
413 CIAP persisted across demographic sensitivity analyses, but did not meet the correction
414 threshold across all nerve module comparisons and weakened after composition adjustment.
415 Genetic candidate expression identified cellular contexts for further study. Because disease
416 and tissue source were partly confounded, the design cannot determine whether apparent
417 differences reflect anatomical compartment, diagnosis or cohort characteristics.
418 The blood analysis separates a broad response in one cohort from directional consistency
419 across cohorts. Six modules met the correction criterion in GSE211225, but no pooled effect
420 was statistically significant. CXCL8 recruitment and complement remained positive across
421 datasets and omission analyses. These directions are compatible with published observations
422 concerning CSF IL8 and leukocyte trafficking, but transcription of a blood module related to
423 CXCL8 is not equivalent to elevated circulating IL8 protein.^{19,36,45,46,47} Small cohorts and
424 differences between mixed blood populations and monocyte pseudobulk limit the strength of
425 conclusions across platforms.

426 The CSF synthesis illustrates the importance of retaining the full eligible reported subset.
427 Including the previously omitted IL6 and CD5 estimates changed the description from eight
428 uniformly concordant proteins to nine concordant proteins among ten. The pooled findings
429 from the model with a fixed effect do not establish independent replication under the source
430 study's regression tests, and high heterogeneity limits interpretation in terms of a common
431 effect.³⁶ Published proteomic and peptidomic findings from PXD002911, together with the
432 three complementary reports, provide descriptive context for inflammation, complement and
433 molecular injury.^{26,33,34,35} Variation in assays and selected reporting prevents treating this
434 evidence map as a comprehensive quantitative synthesis of proteomic findings.

435 CSF and the BNB represent anatomically related but distinct environments. CSF reflects the
436 milieu surrounding nerve roots and the neuraxis, whereas the BNB is formed by endoneurial
437 microvascular endothelial cells within peripheral nerve. The findings support a working
438 model in which endothelial activation at the BNB may influence the entry and local
439 organization of circulating immune effectors. Their temporal relationship to systemic
440 recruitment, CSF inflammation and tissue remodeling remains unresolved because the
441 compartments were sampled in independent cohorts. Paired blood and CSF samples,
442 endothelial permeability assays and spatial profiling of nerve tissue will be needed to
443 determine whether these processes are linked within individual patients.

444 The nerve results distinguish two types of measurement: relative cell representation and
445 transcription within annotated cell groups. Absence of a difference in cell fractions after
446 correction does not exclude altered composition, while a lower macrophage Fc receptor score
447 relative to CIAP does not establish generalized immune suppression. The published C5b-9
448 deposition finding in 52 of 55 CIDP biopsies provides independent histological context for
449 complement at endoneurial capillaries.³⁷ It does not replicate the current transcriptomic
450 contrast or demonstrate that the same abnormalities occur together in individual patients.

451 Peripheral immune profiling in CIDP also reports B cell and CD8 effector changes,
452 emphasizing the relevance of measurements for individual cell types.²⁹
453 None of the intermodule or disability correlations survived correction. The observed positive
454 association between scores in Schwann cells in the repair and damage state and INCAT
455 disability could reflect repair demand, variation in cell state related to injury or other clinical
456 factors. It does not establish repair failure, exhaustion or signalling between endothelial and
457 glial cells. Paired measurements, spatial profiling and perturbation experiments would be
458 needed to test these alternatives.

459 The genetic component adds expression context to published candidate selection.³² *CDH4* and
460 *GNG7* localization can inform the choice of cells for experiments, but expression alone does
461 not identify the effector gene at a locus or the direction of a causal mechanism. The locus
462 finding from the source study's analysis of women should also not be generalized to all CIDP
463 without qualification. Sparse *DIRAS1* and *SLC39A3* expression adds uncertainty, and donor
464 bootstrapping cannot establish function. Independent genetic analyses were not performed
465 because the complete custom CIDP summary statistics were available only on request in the
466 source study. These candidates therefore remain priorities for validation, rather than
467 established therapeutic targets.

468 **Limitations**

469 Several limitations constrain inference. The small blood cohorts produced wide pooled
470 intervals, and effects from whole blood, leukocytes and monocytes are not directly
471 interchangeable. CSF synthesis used selected published coefficients, rather than the full
472 assayed proteome or individual samples. Pooled tests based on a normal approximation differ
473 from the source regressions, and correction across ten reported proteins cannot remove
474 selection bias in the source report. The CSF evidence map based on four studies was not
475 systematic.

476 The nerve comparison used nine CIDP donors and a disease comparator, with eligible
477 numbers varying by cell group. All CIDP donors were male and recruitment centres were
478 imbalanced. Demographic sensitivity analyses reduce concern about those measured factors
479 but cannot establish generalizability to women or remove residual confounding by treatment
480 and clinical differences. Macrophage subtype analyses had few eligible donors. Composition
481 adjustment attenuated the Fc association and increased uncertainty. Composition may reflect
482 disease biology as well as confounding, so this adjustment does not prove that composition
483 explains the original association. The original Fc result did not meet the threshold after
484 correction across all 80 nerve module comparisons. New component and sensitivity analyses
485 were selected after that result was known.

486 An audit of the companion spatial dataset GSE285984 found 99 measured biological genes,
487 with no coverage of the specified Fc module. Only two CIDP and two CIAP spatial donors
488 were included, from the same atlas. These data cannot directly validate the Fc association.
489 The inspected PRIDE deposit and article supplement did not supply a plain donor protein
490 abundance table suitable for immediate reanalysis. Additional CIDP blood datasets were
491 recorded as candidates, but no new analysable external blood matrix was obtained.

492 Consequently, the present revision adds sensitivity analyses and measurement audits without
493 independent external replication.

494 Most importantly, blood, CSF and nerve were sampled independently, and diagnosis was
495 partly linked to sample type. The analysis cannot establish communication within individual
496 patients, temporal progression from GBS to CIDP or causal mediation by the BNB. Genetic
497 estimates were taken from the source publication. Bootstrap intervals quantify sampling
498 variation in expression summaries, not gene function. The public reproducibility materials
499 contain aggregate results and scripts. Complete recalculation requires original inputs and, for

500 genetic localization, a precomputed intermediate containing summaries for individual donors.
501 No new tissue or functional experiments were performed.

502 **Conclusions**

503 Independent public datasets provide complementary views of gene expression related to
504 inflammation and repair in GBS and CIDP. A macrophage Fc module association is
505 consistent across several demographic sensitivity analyses but remains exploratory because of
506 broader multiplicity, composition sensitivity and limited subtype sampling. Pooled blood
507 effects and nerve correlations remain uncertain. Matched independent cohorts, paired
508 biospecimens and functional assays are required before mechanistic or therapeutic
509 conclusions can be drawn.

510 **Data availability**

511 The public databases, accession identifiers, original input filenames and access URLs are
512 listed in Supplementary Data 1. That workbook contains aggregate data underlying the figures
513 and tables, new sensitivity results and an external data feasibility audit. Supplementary
514 Methods describes processing and requirements for reconstructing intermediates for
515 individual donors. Original acquisition dates absent from the preceding archive have not been
516 retrospectively assigned. Files newly acquired for the present nerve reaggregation and spatial
517 coverage audit have actual download dates and checksums in the new input manifest.
518 The public analysis release accompanying this revision (v2.2.0) is archived at
519 <https://doi.org/10.5281/zenodo.22636223> and maintained at
520 <https://github.com/applepie567/gbs-cidp-blood-nerve-barrier>. It includes the corrected ten
521 protein CSF synthesis, new Fc sensitivity analyses, annotation reaggregation and external data
522 audit. Public files contain aggregate results and reconstruction scripts. Complete recalculation
523 of genetic localization additionally requires the intermediate described in Supplementary
524 Methods.

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533 **Competing interests**

534 The authors declare that they have no competing interests.

535 **Author contributions**

536 YF conceived the study, developed the analytical framework, performed the analyses and
537 drafted the manuscript. FF curated data, validated results and revised the manuscript. SXM
538 contributed to validation, interpretation and manuscript revision. SMF contributed to data
539 curation, literature evaluation and manuscript revision. CJ supervised the methodology and
540 critically revised the manuscript. YH supervised the study, coordinated the project and
541 critically revised the manuscript.

542 **Supplementary material**

543 Supplementary Data 1 contains aggregate source data, new sensitivity results and access
544 documentation. Supplementary Methods describes reproducibility and evidence attribution.
545 Supplementary Figures 1 to 4 accompany the article.

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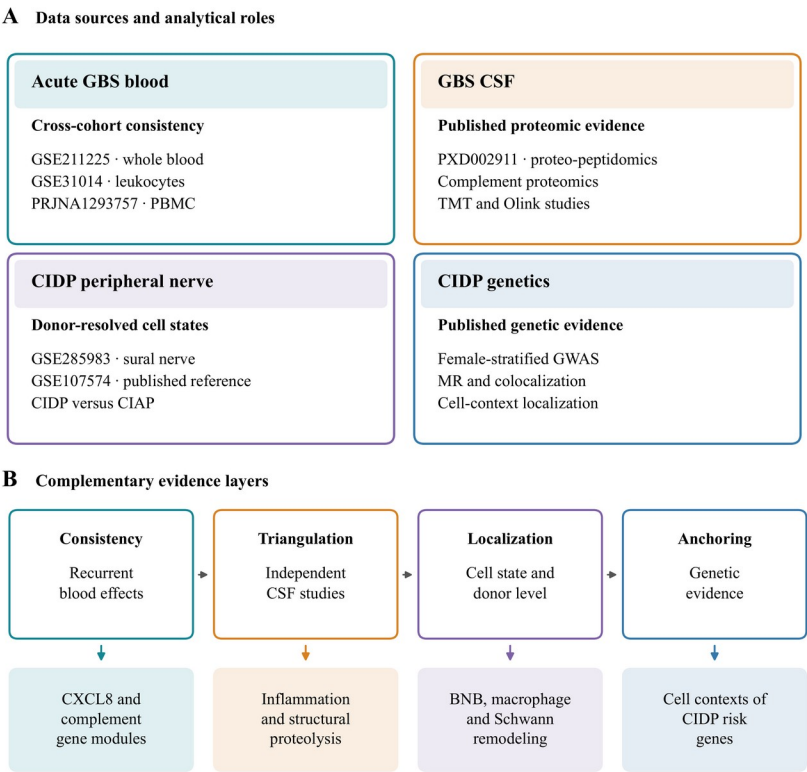
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692 **Figures and legends**



693

694 Figure 1. Study design and integrated model of immune compartments. Published blood, CSF,
695 peripheral nerve and genetic resources were used to compare blood cohorts, integrate CSF findings,
696 analyze nerve responses for individual donors and localize expression of published genetic candidates.
697 The model relates systemic recruitment involving CXCL8 and complement to CSF inflammation and
698 proteolysis, responses of BNB endothelial cells, macrophages and Schwann cells, and the cellular
699 distribution of CIDP susceptibility genes.

700 Alt text: Diagram of independent blood, CSF, peripheral nerve and genetic resources, linked to their analytical
701 roles and proposed biological questions.

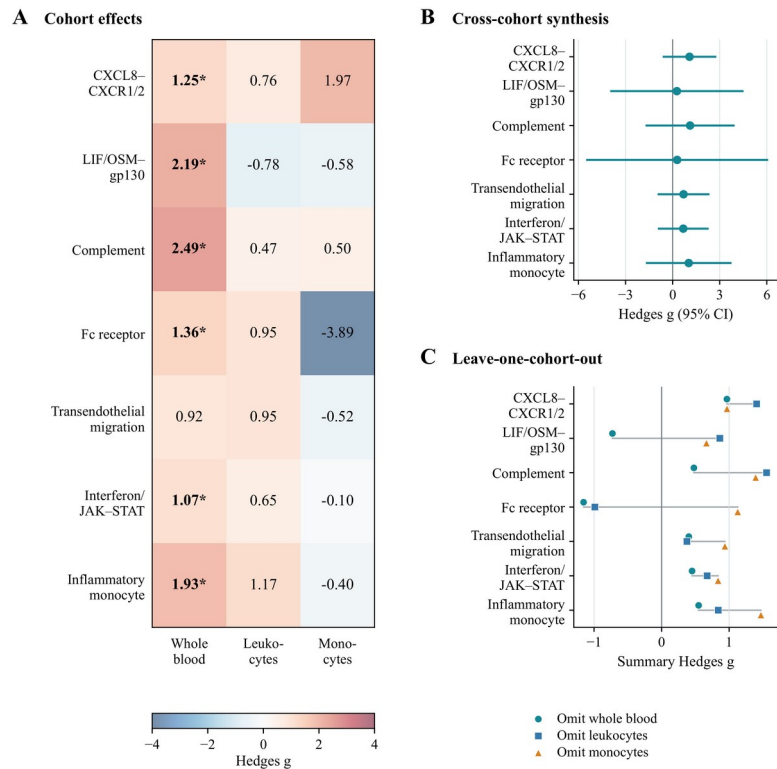
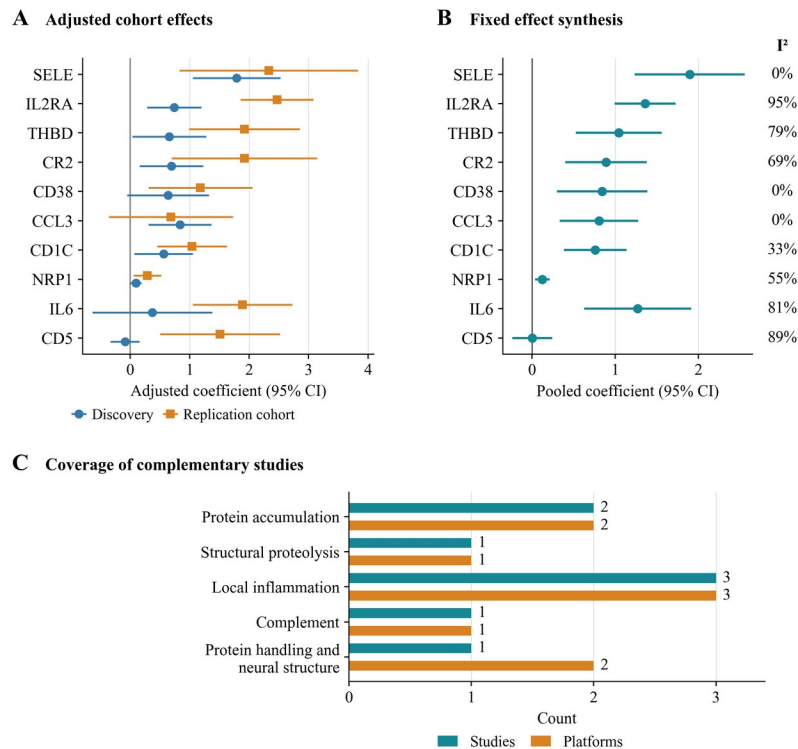


Figure 2. Acute GBS blood programs across cohorts and sensitivity analyses. (A) Hedges g values for seven gene programs in whole blood, leukocytes and monocyte pseudobulk. The programs were defined in the analysis scripts. Asterisks indicate $q < 0.05$ within each cohort. (B) Estimates from models with random effects and 95% confidence intervals calculated using the modified Hartung and Knapp method. (C) Summary effects after each cohort was omitted in turn. CXCL8 signaling through CXCR1 and CXCR2 and the complement gene module retained positive summary effects in all three subsets, with positive individual effects in both retained cohorts. These sensitivity analyses assess stability of direction and do not constitute confirmatory significance tests.

Alt text: Heatmap and forest plots of seven blood gene modules. CXCL8 recruitment and complement effects are positive in all cohorts, while all pooled confidence intervals include zero.



713

714 Figure 3. Synthesis of published CSF estimates. (A) Coefficients from the independent discovery and
 715 replication cohorts of Kmezic et al., adjusted for age, sex and sample handling. Error bars are 95%
 716 confidence intervals calculated from the reported standard errors using a normal approximation.³⁶ (B)
 717 Synthesis using a model with a fixed effect and weighting by inverse variance for all ten CSF proteins
 718 other than IL8 with paired estimates in source Table 4B. Nine pooled effects have $q < 0.05$ after
 719 correction across ten tests. The pooled effect for CD5 does not meet this threshold. I² summarizes
 720 heterogeneity. Pooled significance is distinct from significance in each source cohort. (C) Numbers of
 721 selected studies and platforms covering each biological category, from four complementary
 722 reports.^{26,33,34,35} These counts describe coverage, not statistical validation.

723 Alt text: Cohort and pooled coefficient plots for ten CSF proteins. Nine proteins have positive coefficients in
 724 both cohorts. CD5 changes direction. The lower panel counts study and platform coverage.

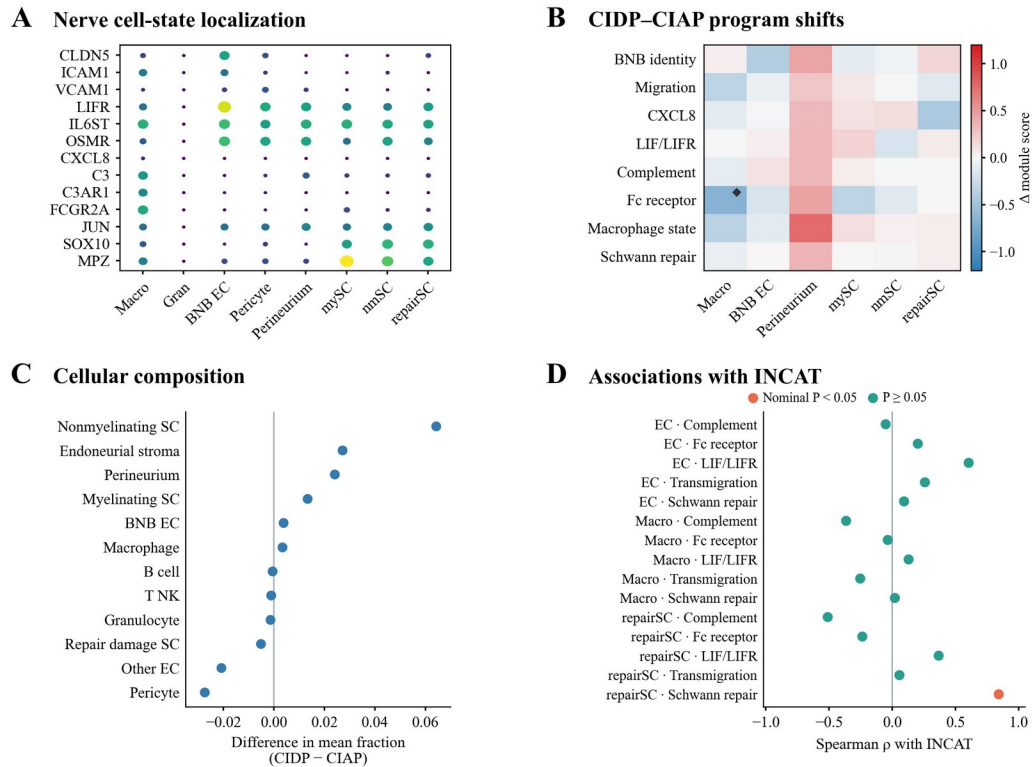
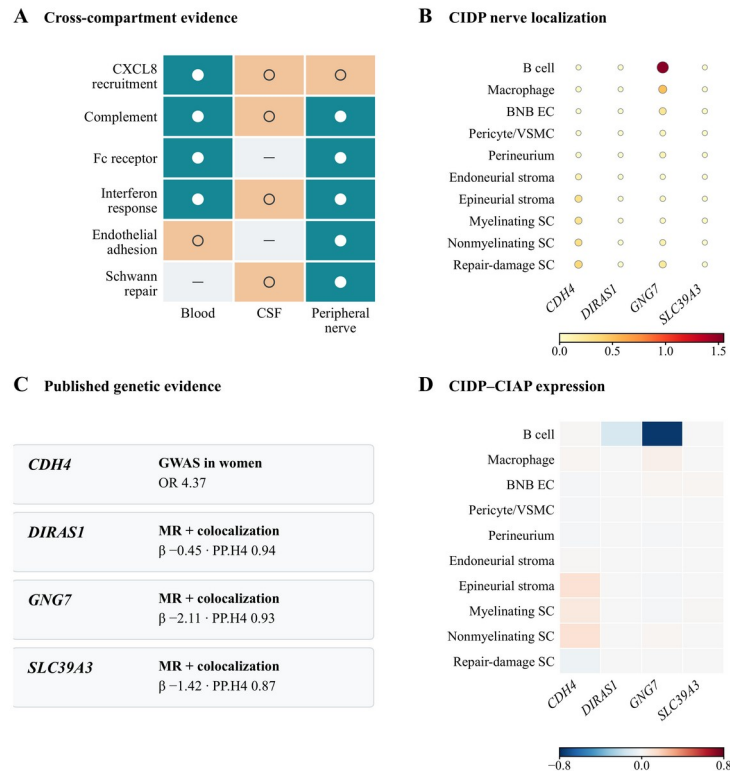


Figure 4. Barrier, immune and repair gene modules in CIDP nerve. (A) Expression across nerve cell states. Dot size indicates detection and color indicates scaled mean expression. (B) Differences in donor gene module scores between CIDP and CIAP. Diamonds indicate $q < 0.05$ within a cell group. No module meets $q < 0.05$ across all 80 eligible comparisons. Supplementary Figures 3 and 4 show the new Fc sensitivity analyses. (C) Differences in mean cell fractions, calculated as CIDP minus CIAP, with donors analyzed as independent samples. Group comparisons in B and C used the U test of Mann and Whitney without covariate adjustment. (D) Spearman associations between donor module scores and INCAT disability. Orange indicates nominal $P < 0.05$ and teal indicates $P \geq 0.05$. None of the associations in D met the threshold after correction for multiple testing. BNB EC refers to endoneurial microvascular endothelial cells.

Alt text: Four panels show nerve gene localization, module score and cell fraction differences, and disability correlations. The macrophage Fc module meets the original correction threshold within macrophages, but not the threshold across all 80 eligible module comparisons. Clinical correlations do not survive correction.



739

740 Figure 5. Immune programs across compartments and cellular localization of genetic candidates in
741 inflammatory peripheral neuropathy. (A) Evidence map across blood, CSF and peripheral nerve.
742 White filled circles indicate direct localization or evidence meeting the threshold after correction for
743 multiple testing, The macrophage Fc classification uses the original correction within a cell group and
744 remains exploratory under broader correction. Open circles indicate recurrent directions or
745 representation in published findings, and dashes indicate programs that were not evaluated. (B)
746 Localization of *CDH4*, *DIRAS1*, *GNG7* and *SLC39A3* across CIDP nerve cell groups, calculated from
747 donor expression summaries in this study. Dot size indicates the percentage of nuclei expressing each
748 gene and color indicates mean $\log_2(CP10k + 1)$. (C) Genetic estimates reported by Du et al.,³² shown
749 to explain candidate selection. These GWAS, Mendelian randomization and colocalization estimates
750 were not recalculated. (D) Differences between CIDP and CIAP for the same genes and cell groups,
751 calculated in this study from donor means of $\log_2(CP10k + 1)$ as CIDP minus CIAP. Red indicates
752 higher expression in CIDP and blue indicates lower expression. None of the displayed comparisons
753 met $q < 0.05$. The color scales represent mean expression in B and differences between diagnoses in
754 D.

755 Alt text: An evidence map is followed by candidate gene expression, published genetic estimates and contrasts
756 between CIDP and CIAP. *CDH4* and *GNG7* have distinct expression patterns. Displayed diagnostic contrasts do
757 not survive correction.

758 **Table 1. Datasets and analytical roles**

Resource	Material	Participants or samples	Analytical role
GSE211225	Whole blood transcriptomics	6 acute GBS, 10 postacute, 6 controls	Primary blood effects for individual participants
GSE31014	Leukocyte microarray	7 GBS, 7 controls	Independent assessment of blood effect direction
PRJNA1293757	PBMC single-cell RNA sequencing	3 AIDP, 2 controls	Monocyte localization and assessment of effect direction
PXD002911	CSF proteomics and peptidomics	GBS (AIDP), MS and controls, with pooled discovery material	Published CSF identifications and functional annotations
GSE285983	Sural nerve single-nucleus RNA sequencing	9 CIDP, 11 CIAP in focused comparison	Donor gene module scores, cell fractions and expression localization
GSE107574	Human endoneurial endothelial transcriptomics	Cultured cells and microvessels	Published BNB context; no independent raw-data reanalysis
Kmezic et al. 2023	CSF Olink proteomics	Discovery: 6 GBS and 20 controls. Replication: 6 GBS and 6 controls	Synthesis of ten paired non-IL8 CSF coefficients from source Table 4B
PXD056286	Sural nerve histology and proteomics	55 biopsies from CIDP or its variants	Published tissue observations and calculated Wilson interval

759 GBS, Guillain–Barré syndrome; AIDP, acute inflammatory demyelinating polyradiculoneuropathy; CIDP, chronic inflammatory demyelinating
760 polyradiculoneuropathy; CIAP, chronic idiopathic axonal polyneuropathy; CSF, cerebrospinal fluid; MS, multiple sclerosis; PBMC, peripheral
761 blood mononuclear cell; BNB, blood–nerve barrier. Cohorts were independently sampled. Focused nerve sample sizes vary with cell-group
762 eligibility.

763 **Table 2. Exploratory synthesis of acute GBS blood gene modules**

Gene module	Summary Hedges g	95% CI lower	95% CI upper	Modified Hartung– Knapp P	I ² , %	Cohorts, n
CXCL8 signaling via CXCR1 and CXCR2	1.075	-0.634	2.785	0.114	0.0	3
LIF and OSM signaling via gp130	0.271	-3.978	4.519	0.810	81.2	3
Complement	1.111	-1.728	3.950	0.234	57.2	3
Fc receptor	0.288	-5.510	6.086	0.851	72.7	3
Transendothelial migration	0.694	-0.958	2.347	0.212	1.1	3
Interferon signaling via JAK and STAT	0.677	-0.943	2.297	0.214	0.0	3
Inflammatory monocyte	1.026	-1.702	3.754	0.247	49.5	3

764 Positive Hedges g indicates a higher acute GBS score. Between-study variance used DerSimonian–Laird estimation; intervals and P values used
765 the modified Hartung–Knapp procedure with residual scale bounded below by one. Each synthesis includes three cohorts. No pooled effect
766 reaches $P < 0.05$. CI, confidence interval.

767 **Table 3. Descriptive evidence map across blood, CSF and peripheral nerve**

Program	Blood evidence	CSF evidence	Peripheral nerve evidence	Interpretation
CXCL8 recruitment	Positive in 3 cohorts; pooled P = 0.114; whole-blood q < 0.05	Published innate immune and inflammatory protein findings	Myeloid localization and endothelial trafficking context	Shared recruitment hypothesis
Complement	Positive in 3 cohorts; pooled P = 0.234; whole-blood q < 0.01	Published complement protein findings	Macrophage localization	Complement-related evidence; amplification remains a hypothesis
Fc receptor	Whole blood q < 0.05, with mixed directions across cohorts	Not directly evaluated	Macrophage q = 0.0087 within eight modules and q = 0.087 across 80 nerve comparisons	Exploratory association. Attenuated after composition adjustment.
Interferon response	Whole blood q < 0.05, with mixed directions across cohorts	Published evidence of antiviral defense	Macrophage, endothelial and Schwann cell states	Response varies by compartment
Endothelial adhesion	Positive blood direction	Not directly evaluated	ICAM1 and VCAM1 localization to BNB endothelial cells	Endothelial context for functional study
Schwann repair	Not evaluated	Published evidence of axonal domain peptide degradation	Repair and stress programs localized to Schwann cells	Proposed link between structural injury and repair

768 q denotes the Benjamini–Hochberg-adjusted P value within the specified testing family. Blood comparisons concern acute GBS versus controls;
769 the nerve comparison concerns CIDP versus CIAP. This map does not constitute a direct paired comparison between compartments or a test of
770 causal mediation.